



**MANIPAL**  
UNIVERSITY COLLEGE  
MALAYSIA

Formerly known as  **MELAKA-MANIPAL**  
MEDICAL COLLEGE  
MANIPAL ACADEMY OF HEALTH EDUCATION

# **MICROSCOPIC STRUCTURE OF SPINAL CORD, CEREBRUM & CEREBELLUM**

Professor Kalyan Chakravarthi Kosuri



● ● ● *#For instruction purpose only..*

23-06-2026

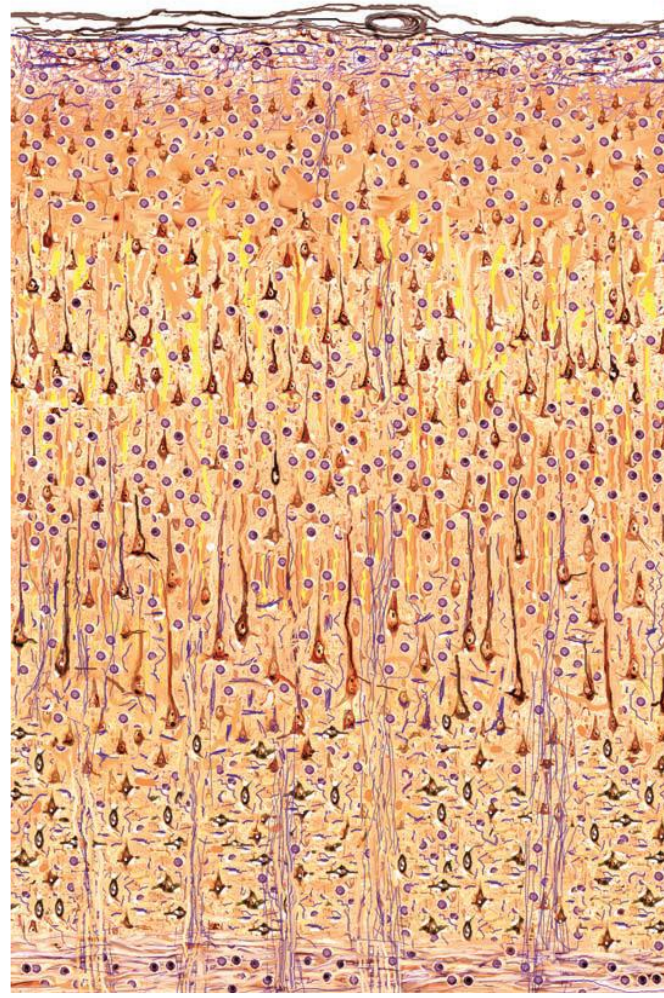
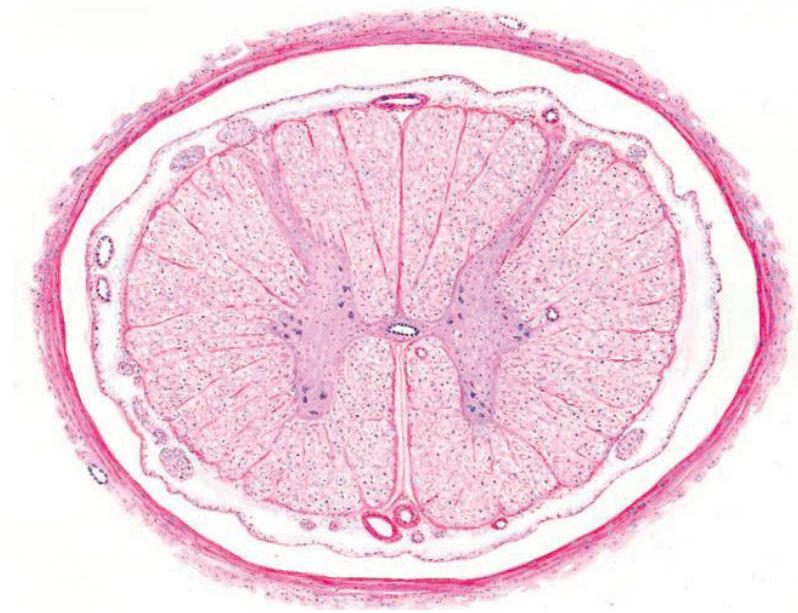
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# LEARNING OBJECTIVES

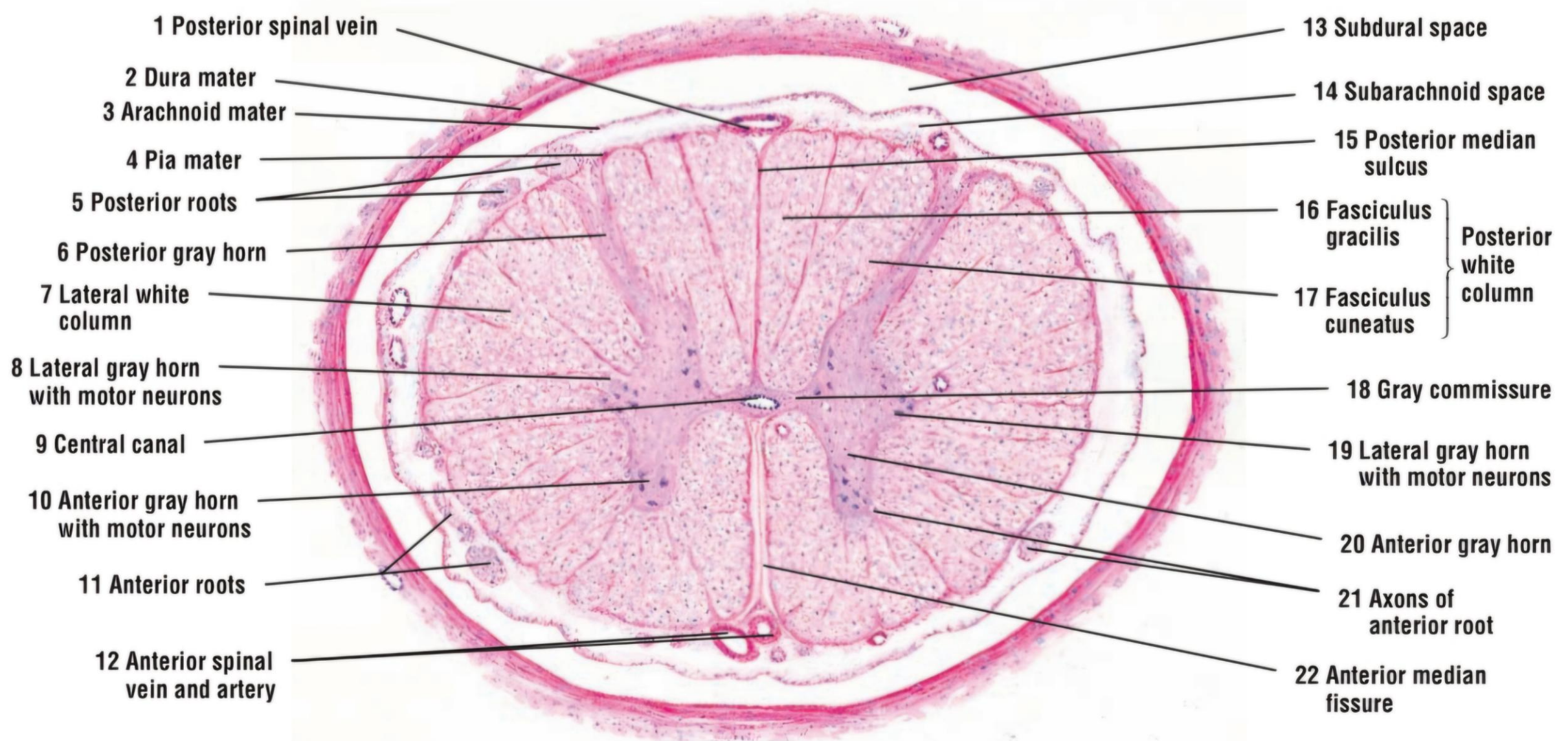
Explain and illustrate the microscopic structure of the following:

1. Spinal cord
2. Cerebrum
3. Cerebellum





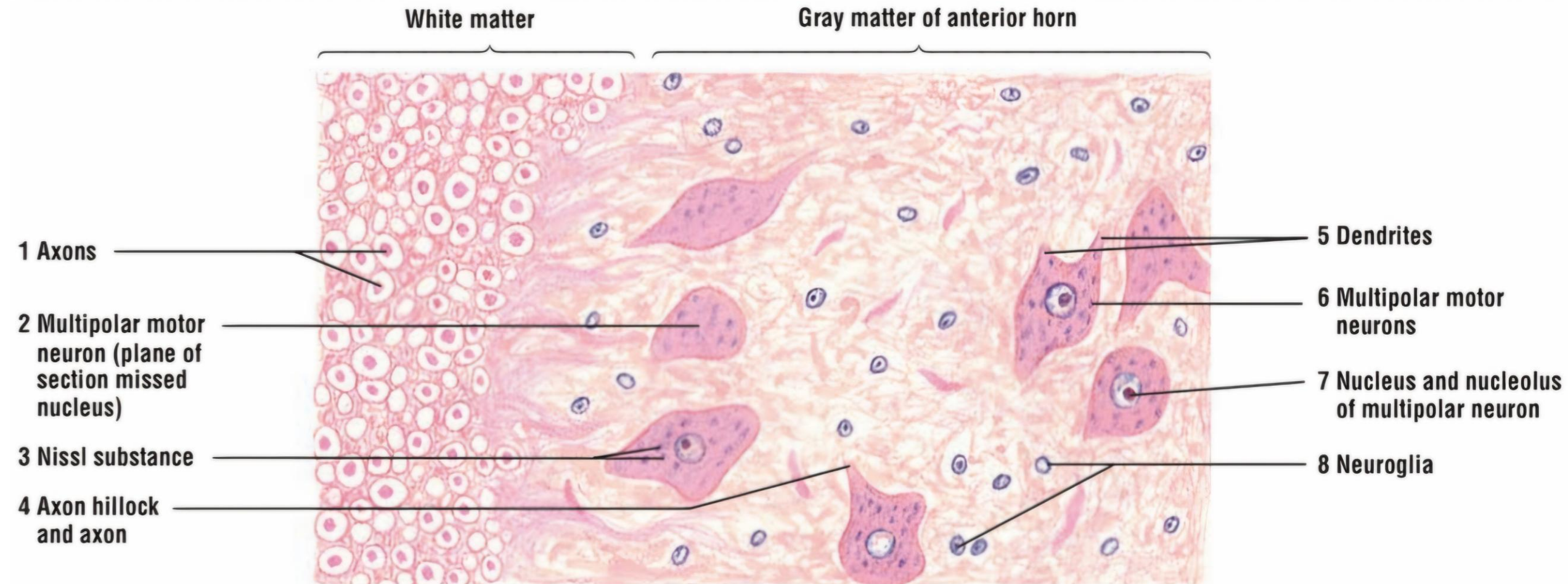
# Microscopic structure of the Spinal Cord: Midthoracic Region (Transverse Section)



Thoracic spinal cord sections show relatively small gray matter, abundant white matter, and a prominent lateral horn.



# Spinal cord: anterior gray horn, motor neuron, and adjacent white matter.



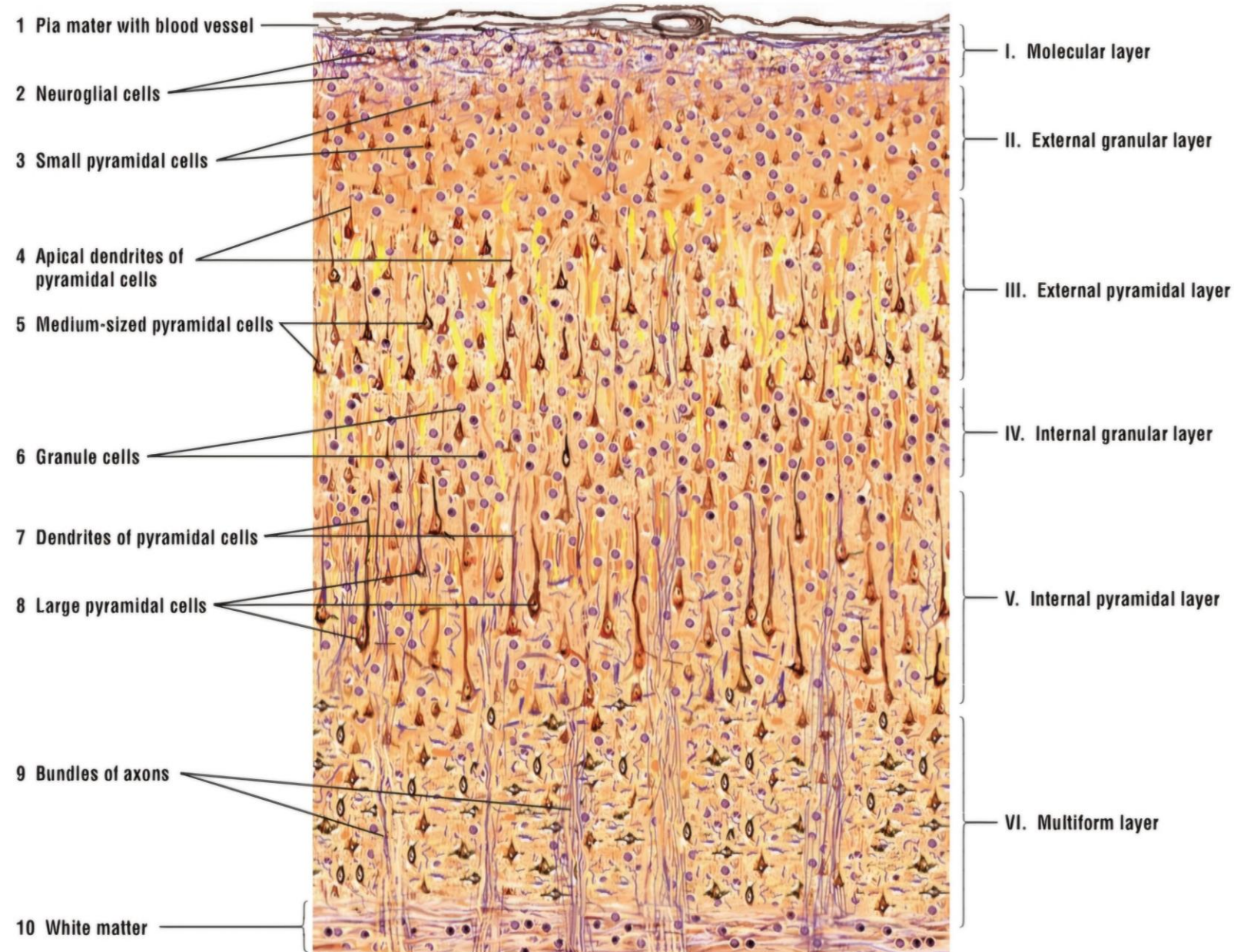
## **Microscopic structure of the Spinal Cord: Midthoracic Region (Transverse Section)**

- Central H-shaped gray matter surrounded by peripheral white matter.
- Anterior (ventral) gray horns contain large multipolar motor neurons.
- Posterior (dorsal) gray horns contain sensory neurons and interneurons.
- Lateral gray horns are present at thoracic levels (T1–L2) and contain sympathetic preganglionic neurons (motor neurons) .
- Central canal is located in the center of the gray commissure.
- Ependymal cells line the central canal and help circulate cerebrospinal fluid (CSF).
- White matter is divided into anterior, lateral, and posterior funiculi containing ascending and descending nerve tracts.
- White matter is more abundant than gray matter at the thoracic level.
- Neuroglial cells (astrocytes, oligodendrocytes, microglia, and ependymal cells) support and protect nervous tissue.
- Characteristic feature: Presence of a distinct lateral horn (intermediolateral cell column), which is a hallmark of the thoracic spinal cord.



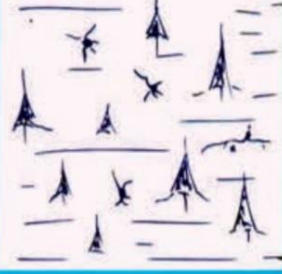
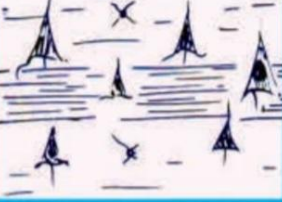
# Microscopic structure of the Cerebral Cortex: Gray matter

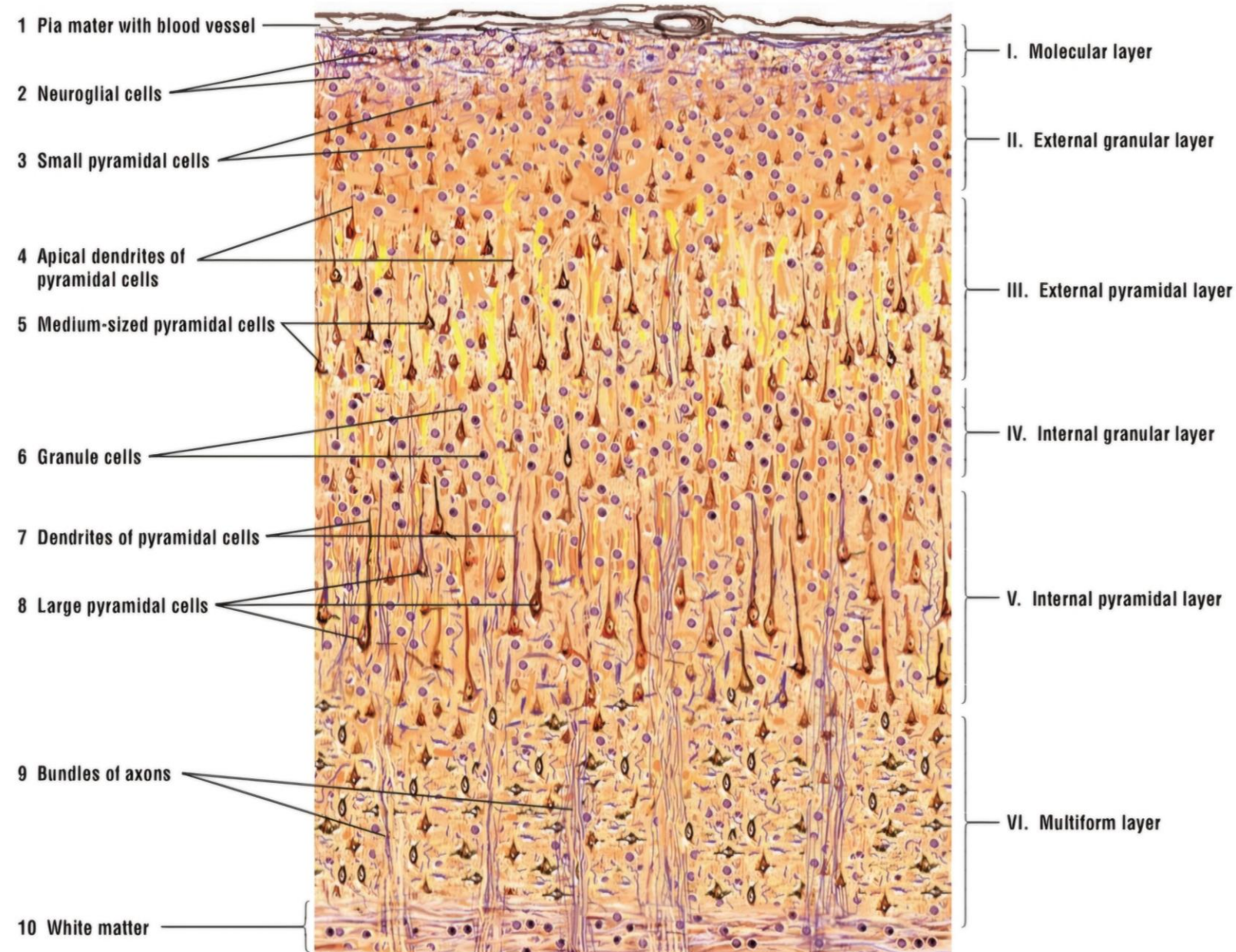
- Outer gray matter (neuronal cell bodies, dendrites, neuroglia, and unmyelinated nerve fibers) and inner white matter (myelinated nerve fibers and supporting neuroglial cells).
- Six-layered cerebral cortex.
- Presence of pyramidal cells and granular cells.
- Abundant neuroglial cells throughout the tissue.





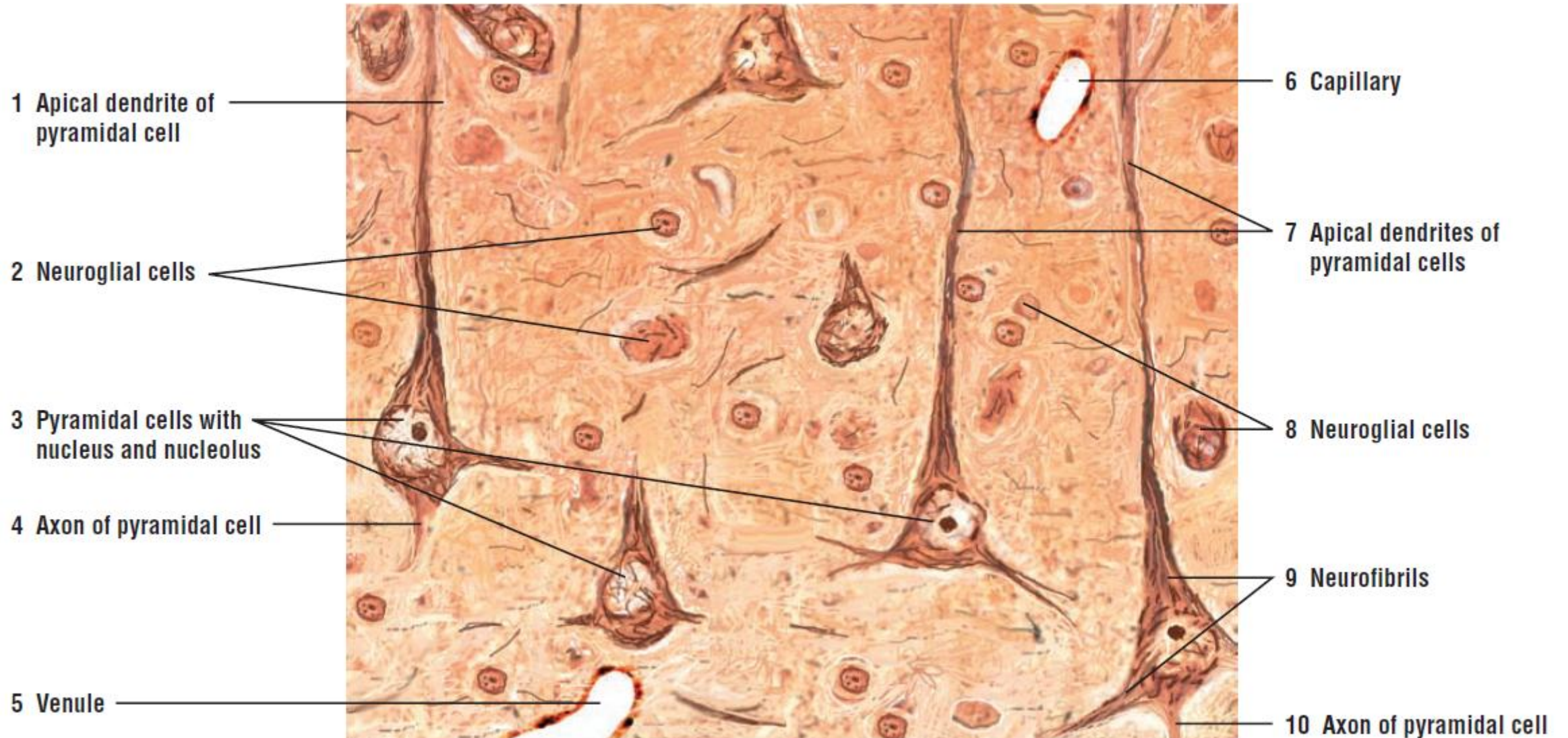
# Microscopic structure of the Cerebral Cortex: Gray matter

|                           |                                                                                     |                                                                        |
|---------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Plexiform or molecular    |    | Transverse fibres and some scattered neurons                           |
| External granular         |    | Mainly stellate neurons                                                |
| Pyramidal                 |    | Mainly pyramidal neurons<br>Some stellate cells and basket cells       |
| Internal granular         |    | Stellate neurons<br>Outer band of Baillarger (myelinated nerve fibers) |
| Ganglionic                |   | Giant pyramidal cells<br>Inner band of Baillarger                      |
| Multiform or polymorphous |  | Neurons of various sizes and shapes<br><br>Merges with white matter    |





## Layer V (Internal pyramidal layer)



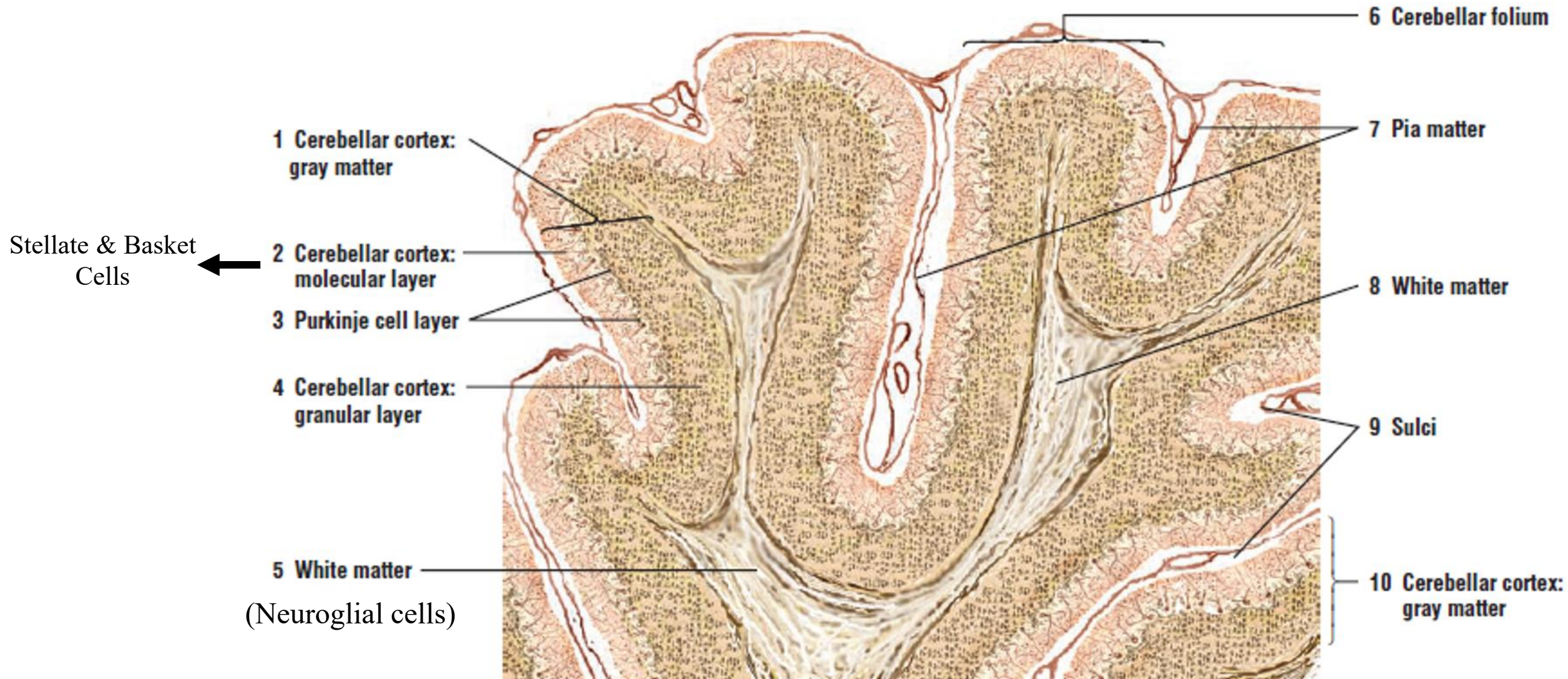


# Microscopic structure of the Cerebral Cortex: Gray matter

- The cerebrum is composed of an outer gray matter (cerebral cortex) and inner white matter.
- The cerebral cortex consists mainly of neuronal cell bodies, dendrites, neuroglia, and unmyelinated nerve fibers.
- The white matter contains myelinated nerve fibers and supporting neuroglial cells.
- The cerebral cortex is typically 2–4 mm thick. The cortex is arranged into six distinct layers.
- Layer I (Molecular layer): Contains few neurons and numerous nerve fibers.
- Overlying and covering the molecular cell layer (I) is the delicate connective tissue of the brain, the pia mater.
- Layer II (External granular layer): Contains mainly different types of neuroglial cells and small pyramidal cells.
- The apical dendrites of the pyramidal cells (4, 7) are directed toward the periphery of the cortex, whereas their axons extend from the cell bases.
- Layer III (External pyramidal layer): Contains small and medium-sized pyramidal cells.
- Layer IV (Internal granular layer): Rich in stellate cells (granule cells) and receives sensory input.
- Layer V (Internal pyramidal layer): Contains largest pyramidal cells, including Betz cells in the motor cortex.
- Layer VI (Multiform/Fusiform layer): Contains neurons fusiform cells, granule cells, stellate cells, and cells of Martinotti. and merges with white matter.
- Neuroglial cells (astrocytes, oligodendrocytes, and microglia) provide support, nutrition, myelination, and defense.

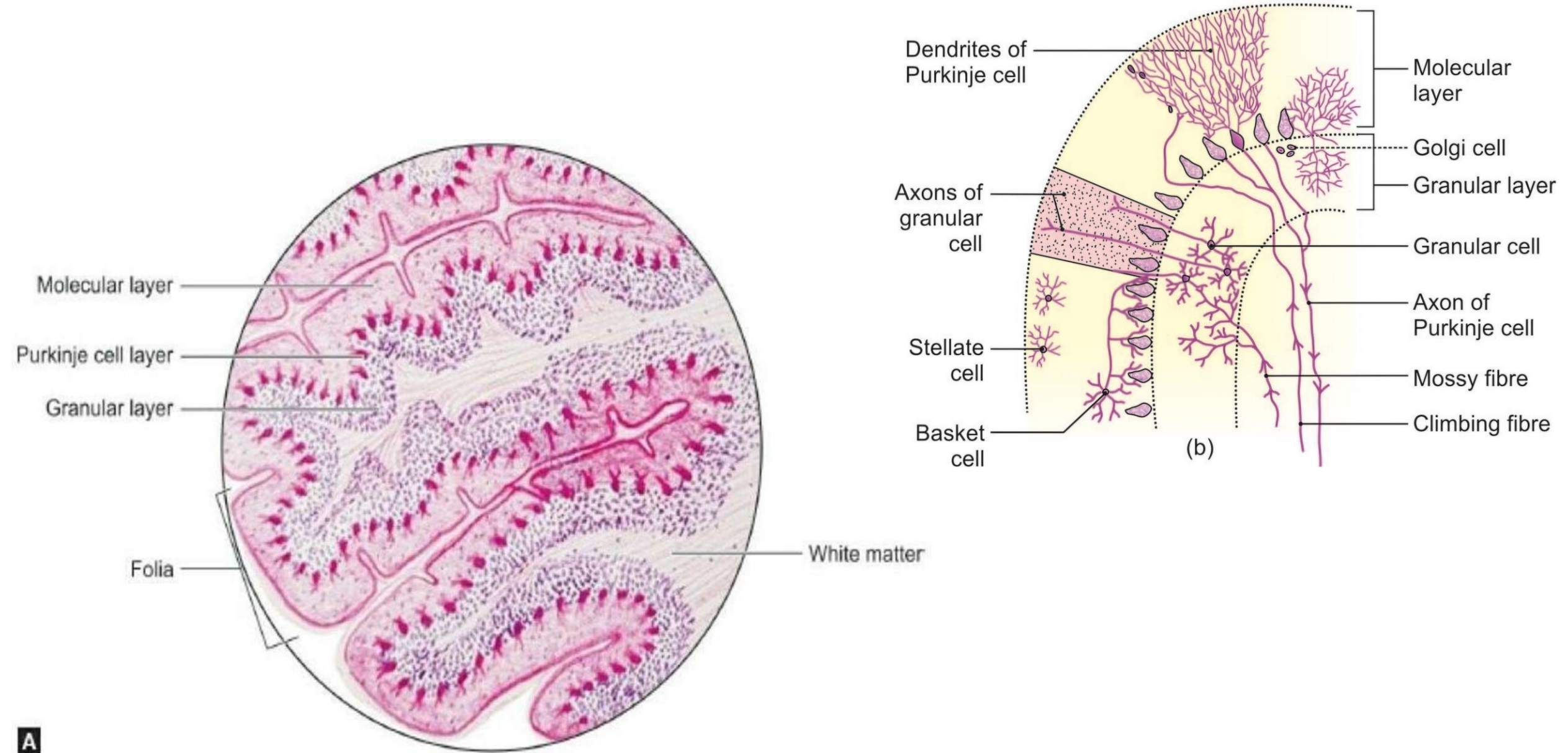


## Microscopic structure of the Cerebellum (Transverse Section)





# Microscopic structure of the Cerebellum (Transverse Section)





## **Microscopic structure of the Cerebellum (Transverse Section)**

- The cerebellum consists of an outer cortex (gray matter) and an inner medulla (white matter).
- Deep folds in cortex called cerebellar folia separated by sulci.
- The cerebellar cortex is arranged into three distinct layers.
- The outer Molecular Layer contains few neurons and numerous nerve fibers.
- The Molecular Layer contains stellate and basket cells, which act as interneurons.
- The middle Purkinje Cell Layer consists of a single row of large flask-shaped Purkinje neurons.
- Purkinje cells are the principal output neurons of the cerebellar cortex.
- The dendrites of Purkinje cells extend into the molecular layer.
- The inner Granular Layer is densely packed with small granule cells.
- Granule cells are the most numerous neurons in the cerebellum.
- The white matter (medulla) contains myelinated nerve fibers and neuroglial cells. Deep cerebellar nuclei are located within the white matter and receive impulses from Purkinje cells.
- Neuroglial cells (astrocytes, oligodendrocytes, and microglia) support and protect cerebellar tissue.

## **THE REFERENCE BOOKS FOR FURTHER READINGS.....**

1. Gray's Anatomy by Churchill Livingstone 42nd Ed.
2. Snell's Clinical Neuroanatomy, 9th Edition.
3. Textbook of Clinical Neuroanatomy-2Edition Vishram Singh.
4. Textbook of Human Neuroanatomy Fundamental and Clinical -  
Inderbir Singh 10th Ed.





THANK YOU